Final Project. Modernizing Deep SORT.

Deep learning for computer vision

HSE, 2025/2026

github.com/ironyoid/deepsort-mod

1. What I did

For this project I took the original Deep SORT tracker and swapped its two main parts — the person detector and the appearance model — for newer ones. I kept the original tracker code itself unchanged and just built around it, so I could plug different detectors and re-ID models in and out and compare them. After that I tuned the parameters, added a separate identity database (the extra task), and tried a segmentation model.

I measure how good the tracking is with HOTA, averaged over the six test videos, using TrackEval (the standard MOT evaluation tool). The original Deep SORT gets 0.40. My best setup gets 0.57, and it beats the original on every single video.

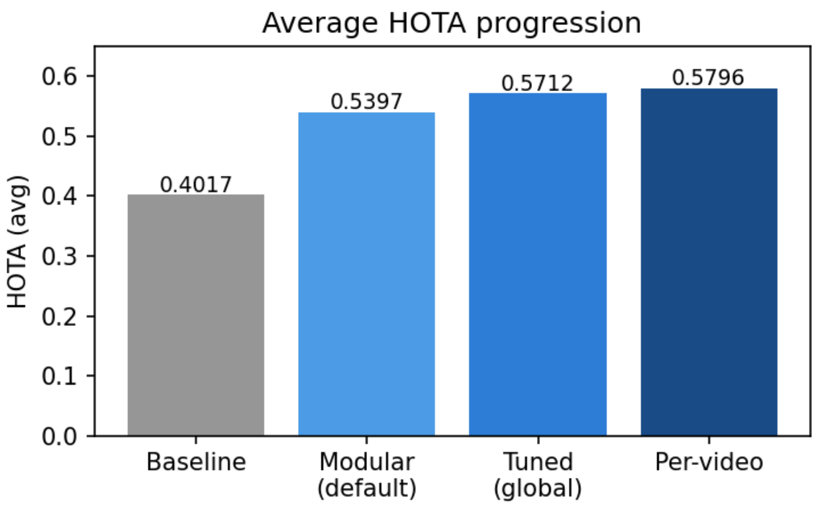

Figure 1: How the average HOTA went up as I improved the system.

2. Datasets

I use six MOTChallenge videos: TUD-Campus, TUD-Stadtmitte, KITTI-17, PETS09-S2L1, MOT16-09 and MOT16-11.

I use three kinds of measurement:

- For the whole tracker: HOTA (averaged over the videos). HOTA mixes two things — how good the boxes are (DetA) and how consistent the IDs are (AssA). I evaluate MOT15 and MOT16 with their own standard settings and average the per-video HOTA.
- For detectors on their own: Precision, Recall and F1 against the ground-truth boxes.
- For re-ID models on their own: clustering scores on ground-truth crops (Silhouette, Calinski-Harabasz, Fowlkes-Mallows), and HOTA using ground-truth boxes so the detector doesn't get in the way.

3. How the system is put together

Every frame goes through three steps: a detector finds the people, a re-ID model turns each box into a 512-number "fingerprint", and the original Deep SORT tracker assigns the IDs. I pick the detector and re-ID model by name before running, so I can test any combination.

4. Baseline

First I measured the original with it's `mars-small128` appearance model. This is what I have to beat.

	TUD-Campus	TUD-Stadt.	KITTI-17	PETS09	MOT16-09	MOT16-11
HOTA	0.3986	0.3675	0.4341	0.4484	0.3624	0.3995

Table 1: Original Deep SORT, HOTA per video. Average = 0.4017.

5. Choosing a detector

I tried three detectors: YOLOv8n, RT-DETR (Ultralytics) and Faster R-CNN (torchvision). That's three models from two different libraries. YOLO came out best, both on detection quality and on the final HOTA.

Detector	Library	Precision	Recall	F1
yolov8n	Ultralytics	0.849	0.728	0.784
rtdetr	Ultralytics	0.645	0.784	0.708
fasterrcnn	torchvision	0.510	0.765	0.612

Table 2: Detector quality against the ground truth (confidence 0.3).

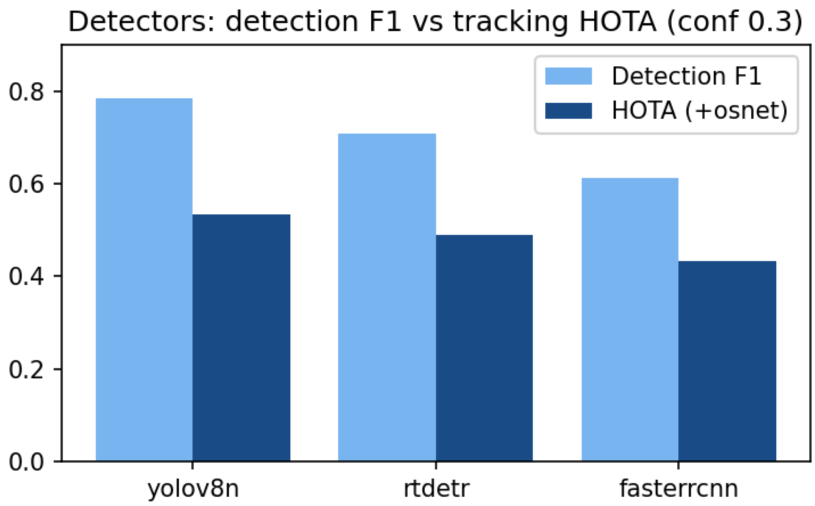

Figure 2: Detection F1 and the final HOTA for each detector (with osnet_x0_5, confidence 0.3).

RT-DETR and Faster R-CNN look weak here, but that's mostly because at confidence 0.3 they produce a lot of false boxes. If I raise the confidence to 0.5, RT-DETR jumps from 0.475 to 0.540 HOTA (about the same as YOLO) and Faster R-CNN goes to 0.466. So the confidence threshold really matters and is worth tuning per detector.

6. Choosing a re-ID model

I tried four OSNet models from Torchreid (`osnet_x0_25`, `osnet_x0_5`, `osnet_x1_0`, `osnet_ain_x1_0`). They are trained on large person datasets, which helps them work on videos they haven't seen.

Re-ID model	HOTA	Silhouette	Calinski-H	Fowlkes-M
osnet_x0_25	0.5335	0.2594	124.6	0.6122
osnet_x0_5	0.5397	0.2744	123.8	0.6406
osnet_x1_0	0.5323	0.2821	130.8	0.6228
osnet_ain	0.5289	0.2796	126.8	0.6380

Table 3: Re-ID models: HOTA (yolov8n, confidence 0.3) and clustering scores on ground-truth crops.

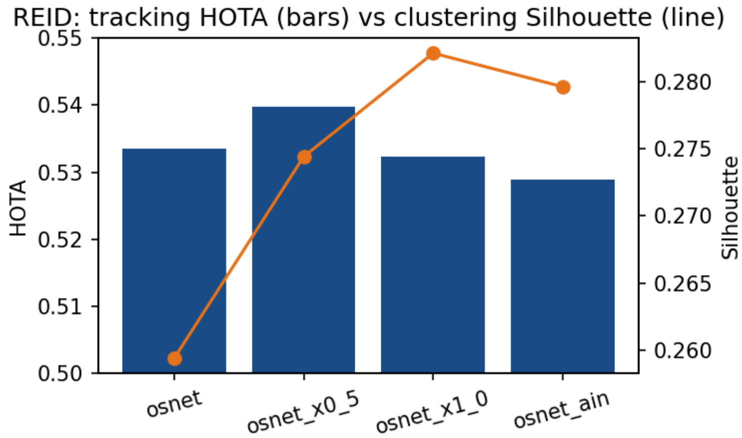

Figure 3: Re-ID HOTA (bars) next to the Silhouette score (line).

One thing surprised me: the metrics don't agree. `osnet_x1_0` has the most separated fingerprints (best Silhouette) and wins when I give the tracker perfect boxes, but `osnet_x0_5` wins in the real pipeline with noisy boxes. When I feed in ground-truth boxes (so only the re-ID matters), HOTA is around 0.85 for all four models — much higher than the 0.55 of the full system. So the detector, not the re-ID model, is what's really holding the system back.

7. Tuning the parameters

I swept the three parameters that matter most. Each one has a sweet spot: if the appearance matching is too strict the IDs break apart, if it's too loose different people get merged, and a larger `max_age` helps a track survive being hidden, but only up to a point.

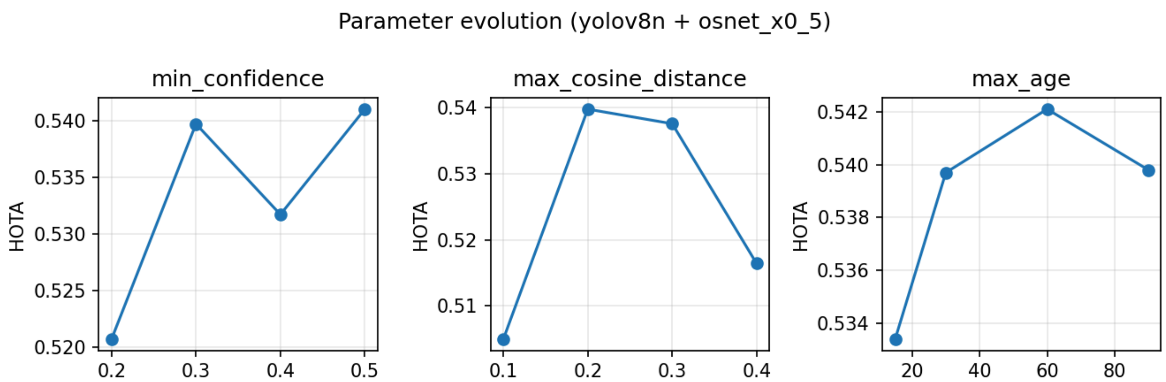

Figure 4: Changing one parameter at a time (yolov8n + osnet_x0_5).

Since the detector is the main bottleneck, I switched to the bigger `yolov8s` and `osnet_x1_0` and ran a grid search. The best single setting was confidence 0.5, max-cosine-distance 0.3, max-age 60, giving HOTA 0.5712. The task also allows different parameters per video, which pushes the average to 0.5796 — busy, long videos like MOT16-11 prefer keeping more detections (lower confidence), while calmer videos do better with a higher confidence.

Video	Baseline	Best (per-video)	Settings
TUD-Campus	0.3986	0.5205	conf 0.5 / mcd 0.4
TUD-Stadtmitte	0.3675	0.6353	conf 0.5 / mcd 0.3 / age 60
KITTI-17	0.4341	0.6037	conf 0.5 / mcd 0.4 / age 30
PETS09-S2L1	0.4484	0.6099	conf 0.5 / mcd 0.2 / age 30
MOT16-09	0.3624	0.5519	conf 0.5 / mcd 0.3 / age 90
MOT16-11	0.3995	0.5565	conf 0.3 / mcd 0.2 / age 60

Table 4: Per-video baseline vs my best settings. Averages: 0.4017 vs 0.5796.

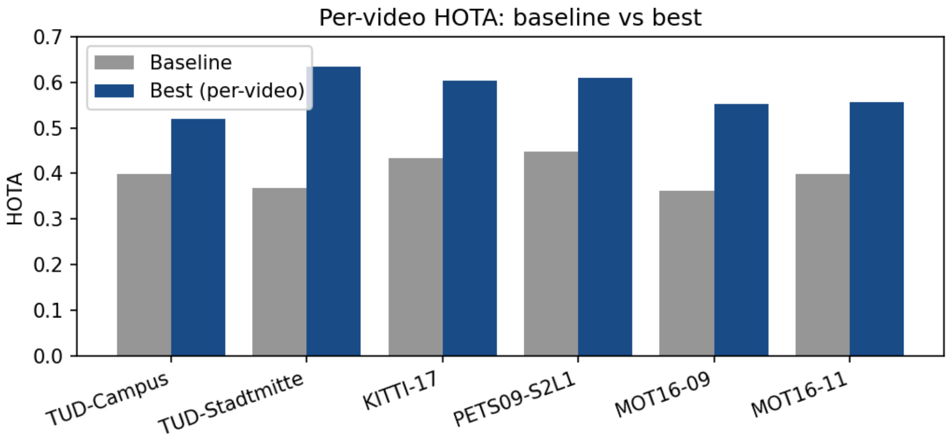

Figure 5: HOTA for each video: baseline vs best.

8. Extra task: an identity database

For the extra task I added a separate "identity" layer on top of the tracker. The idea is simple: a track ID is short-lived (if someone disappears for a while and comes back, the tracker gives them a new ID), but an identity is meant to be permanent. I keep a small database of identities so that a person who leaves and returns gets recognized as the same person again.

Each frame I detect, get the fingerprints, run the normal tracking step, and then for every active track I look up its fingerprint in the database with nearest-centroid kNN (cosine distance with a threshold). If it's close enough to a known identity I use that one, otherwise I create a new identity. Each track keeps a short history of identity guesses, and over a time window I take the majority vote. If two tracks end up claiming the same identity at the same time (which can't be right) I reset them. I reuse the tracker's OSNet fingerprints and store each identity as the running average of its fingerprints.

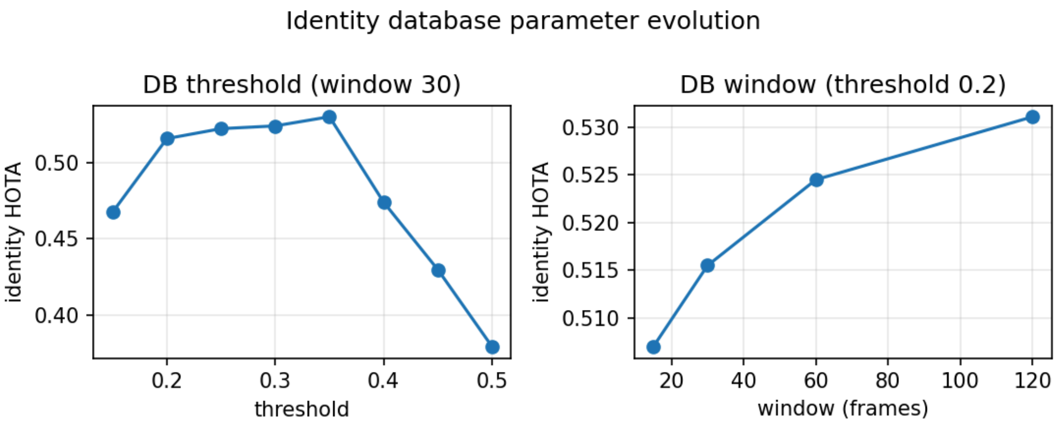

Figure 6: Tuning the database threshold and the voting window.

Same story as before: if the threshold is too strict I get too many identities, too loose and I merge people, and a longer voting window is more stable. The best setting (threshold 0.35, window 180) gives an identity HOTA of 0.5348, and the number of identities per video looks reasonable (about 10–12 for the TUD videos, 19 for PETS09). This is a bit lower than the track-ID HOTA, which makes sense — keeping a global identity is harder than keeping a short-term track ID — but the database does something the plain tracker can't: it recognizes people again after they've been gone.

9. Segmentation

I also added a segmentation model (YOLOv8-seg) that I can switch in instead of a normal detector. It gives a mask for each person; I turn the mask into a box for the tracker, and I can also use the mask to black out the background of each crop before the re-ID step.

Setup	Detection F1	HOTA	Note
yolov8s + osnet_x1_0	0.78	0.5712	reference (boxes)
yolov8s-seg + osnet_x1_0	0.80	0.5703	boxes from masks
yolov8s-seg + masked re-ID	0.80	0.5548	background removed

Table 5: Segmentation results.

The segmentation model works about as well as the normal detector (HOTA 0.5703, slightly higher detection F1). But masking the background for re-ID actually made things a little worse. I think this is because OSNet was trained on normal crops that include the background, so giving it a black background confuses it more than it helps. Not the result I expected, but a useful one.

10. Conclusion

Overall, swapping in modern models and tuning them raised the average HOTA from 0.4017 to 0.5712 with one setting, or 0.5796 with per-video settings — and it beats the original on every video while still running in real time.

- My best setup is yolov8s + osnet_x1_0 with confidence 0.5, max-cosine-distance 0.3, max-age 60.
- The detector matters most. With perfect boxes the HOTA is around 0.85, so the detector is the main limit, not the re-ID model.
- The clustering scores and the actual HOTA don't always agree, so it's worth checking both before picking a re-ID model.
- The identity database (extra task) adds long-term recognition, and the segmentation model also works in the pipeline.

11. How to reproduce

The six videos go under `data/sequences/`; the model weights download automatically the first time. Main commands:

```
# baseline
python tools/generate_detections.py --model=resources/networks/mars-small128.pb \
    --mot_dir=data/sequences --output_dir=data/detections
python evaluate_motchallenge.py --mot_dir=data/sequences --detection_dir=data/detections
\
    --output_dir=results/baseline --min_confidence=0.3 --nn_budget=100
python eval_hota.py --results_dir=results/baseline

# best setup
python run_tracker.py --mot_dir=data/sequences --output_dir=results/best \
    --detector=yolov8s --reid=osnet_x1_0 --min_confidence=0.5 --max_cosine_distance=0.3
--max_age=60
```

```
# evaluating parts, tuning, identity database, segmentation
python eval_detector.py --mot_dir=data/sequences --detector=yolov8n --min_confidence=0.3
python eval_reid.py --mot_dir=data/sequences --reid=osnet_x1_0
python tune.py --mot_dir=data/sequences --detector=yolov8s --reid=osnet_x1_0 \
    --min_confidence=0.3,0.4,0.5 --max_cosine_distance=0.2,0.3,0.4 --max_age=30,60,90
python run_identity.py --mot_dir=data/sequences --output_dir=results/identity \
    --detector=yolov8s --reid=osnet_x1_0 --threshold=0.35 --window=180
python run_tracker.py --mot_dir=data/sequences --output_dir=results/seg \
    --detector=yolov8s-seg --reid=osnet_x1_0 --mask
```